A survey on large language model based autonomous agents

Automated LLM-assisted Review Pipeline

Abstract

Large language model (LLM) based autonomous agents have emerged as a transformative paradigm in artificial intelligence, integrating reasoning, memory, planning, tool use, and multi-agent coordination. This survey systematically synthesizes findings from 187 included studies, drawn from 5,271 downloaded records, and organized across eight core thematic areas derived from the literature (studies may belong to multiple themes). The review identifies dominant cross-study patterns: a consistent advocacy for unified, modular agent architectures that decouple core components; the integration of memory modules with chain-of-thought reasoning to enhance task completion across sequential and cooperative settings; and the growing deployment of LLM-based agents in social simulation, robotics, autonomous driving, and natural science applications. A notable contrast emerges between memory-augmented approaches that achieve reinforcement-learning-comparable efficiency with minimal storage and richer frameworks that yield substantial gains through extensive episodic buffers. Evaluation trends reveal a shift from qualitative demonstrations toward quantitative benchmarks (e.g., BLEU, F1, success rates), yet many surveys note that systematic evaluation remains underreported. Persistent challenges across themes include limited context windows, agent brittleness, difficulty maintaining up-to-date knowledge, and the absence of standardized benchmarks for multi-step reasoning and real-world alignment. By mapping architectural designs, application domains, evaluation strategies, and open problems, this survey provides a comprehensive foundation for advancing LLM-based autonomous agents.

1 Introduction

1.1 Background

Large language models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation, enabling a new class of artificial intelligence systems known as LLM-based autonomous agents. These agents integrate LLMs as central reasoning cores with additional modules for memory, planning, and action, allowing them to perceive environments, form plans, execute tasks, and learn from feedback in an autonomous manner. The convergence of advances in LLMs and agent architectures has accelerated progress toward human-level intelligence, fueling a surge of research across disciplines such as social science, natural science, and engineering. As the field expands rapidly, a growing body of work proposes diverse designs, application scenarios, and evaluation methodologies, making a systematic synthesis increasingly valuable.

1.2 Motivation and gap

Despite this flurry of activity, the literature on LLM-based autonomous agents remains fragmented. Individual studies often focus on narrow aspects—construction details of a specific agent, a single application domain, or a particular evaluation metric—without offering an integrated perspective. Existing surveys tend to cover only subsets of the field, such as agent architectures or applications, and frequently lack a unifying framework that connects design choices to downstream performance.

Furthermore, the evaluation landscape is scattered, with no consensus on standardized benchmarks or protocols. This fragmentation hinders researchers and practitioners from obtaining a holistic understanding of the state of the art, identifying systematic patterns, and recognizing open challenges. Thus, a comprehensive review that synthesizes construction, application, and evaluation within a coherent structure is needed.

1.3 Review objective

This survey aims to provide a comprehensive and systematic review of LLM-based autonomous agents from a holistic perspective. Concretely, we first propose a unified framework that captures the essential components of agent construction, encompassing memory, planning, action, and their interplay with the LLM core. We then present a broad overview of the diverse applications of these agents across social science, natural science, and engineering, highlighting common themes and domain-specific adaptations. Finally, we examine the evaluation strategies commonly employed, discussing their strengths, limitations, and the emerging need for standardization. Drawing on this synthesis, we identify key challenges and outline promising future directions for the field.

1.4 Research questions

To fulfill the review objective, we address the following research questions:

1. How are LLM-based autonomous agents constructed? In particular, what are the main architectural components and design choices?

2. What are the applications of LLM-based autonomous agents in social science, natural science, and engineering? Which tasks and domains have been explored?

3. How are LLM-based autonomous agents evaluated? What metrics, benchmarks, and evaluation protocols are used, and what are their shortcomings?

1.5 Contributions

The main contributions of this survey are as follows:

- We provide a comprehensive systematic review that integrates the construction, application, and evaluation of LLM-based autonomous agents into a single coherent narrative.

- We propose a unified agent framework that synthesizes existing architectural designs, facilitating comparison and future development.

- We offer a structured overview of applications across multiple disciplines, revealing patterns of use, domain-specific challenges, and underexplored areas.

- We analyze current evaluation strategies, identifying common practices and gaps, and discuss implications for reproducibility and comparability.

- We delineate open challenges and promising research directions to guide future work in this rapidly evolving field.

2 Methods

2.1 Review design and objective

This systematic review provides a comprehensive survey of studies on large language model (LLM) based autonomous agents. The objective is to examine the construction, applications, and evaluation of such agents from a holistic perspective. A unified framework is proposed to encompass prior work, followed by an overview of applications across social science, natural science, and engineering, and a synthesis of commonly used evaluation strategies. The review also identifies challenges and future directions in the field.

2.2 Search sources and keyword strategy

Searches were conducted in five bibliographic databases: OpenAlex, Crossref, Semantic Scholar, Europe PMC, and PubMed. The keyword strategy combined terms related to large language models and autonomous agents, including ‘‘large language model’’, ‘‘autonomous agent’’, ‘‘human-level intelligence’’, ‘‘LLM-based autonomous agents’’, ‘‘LLM agent’’, ‘‘agent architecture’’, ‘‘memory module’’, ‘‘planning’’, ‘‘action module’’, ‘‘LLM-based agent’’, ‘‘agent construction’’, ‘‘agent application’’, and ‘‘evaluation’’, as well as the publication types ‘‘survey’’, ‘‘systematic review’’, and ‘‘review’’. The search was limited to English-language records published between January 2021 and August 2023.

2.3 Eligibility criteria

Studies were included if they addressed large language models in the context of autonomous agents, including surveys, reviews, or original research on agent construction, application, or evaluation. No formal exclusion criteria were applied; all retrieved records were considered for screening.

2.4 Article retrieval, cleaning, and ranking

A total of 5,271 records were downloaded from the five databases. After deduplication and cleaning based on Digital Object Identifier (DOI) metadata, 4,962 unique records remained. These records were ranked using a relevance scoring strategy based on title and abstract content, producing a ranked list of 4,962 records. The top 1,000 records from this ranking (the *top-ranked screening pool*) were selected for subsequent screening. From this pool, the LLM-based screening process targeted the top 200 records for full-text retrieval and evidence extraction.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text files were successfully obtained for all 200 targeted records, comprising PDFs, text-like files, XML/TEI, JSON, and other formats. An LLM-assisted pipeline generated evidence cards for each record, extracting structured information on study characteristics, findings, and relevance. The screening decisions yielded 187 included records, 11 excluded records, and 2 uncertain records; all 200 evidence cards were produced using full-text content. The chunk count distribution of the output was as follows: 175 records were processed as a single chunk, 24 records required six chunks, and 1 record required seven chunks.

2.6 Evidence-card completeness and synthesis preparation

An evidence-card completeness and extraction audit was conducted to verify the consistency and coverage of the extracted fields. No formal quality scores were assigned; the audit focused on the completeness of the evidence cards and the accuracy of extracted information. The 187 included cards formed the basis for the narrative synthesis and thematic analysis presented in this review. Studies were permitted to belong to multiple themes, and theme counts reflect overlapping assignments.

3 Results

3.1 Study selection flow

The search and screening process followed a structured multi-stage flow. From an initial download of 5,271 records, 4,962 remained after DOI metadata cleaning and duplicate removal. All cleaned records were ranked by relevance based on abstract and title content, and the top 1,000 records were designated as the screening pool. From this pool, the 200 highest-ranked articles were selected for LLM-based article card generation. After LLM screening, 187 articles were included for synthesis, 11 were excluded, and 2 were classified as uncertain; no decisions were missing. The overall selection flow is summarised in Table **??** and illustrated in the PRISMA flow diagram (Figure **??**).

3.2 Candidate metadata and ranking coverage

The initial 5,271 records were obtained from database retrieval. After cleaning, 4,962 records retained sufficient metadata (DOI, title, abstract) for ranking. Relevance scores based on abstract and title similarity to the review objective were calculated, and the top 1,000 records were retained as the screening pool. From this pool, the 200 most relevant records were advanced to full-text acquisition and article card generation, ensuring that the highest-ranked studies received priority for detailed evidence extraction.

3.3 Full-text availability

Full-text documents were successfully obtained for all 200 candidate articles. The full-text corpus comprised 71 PDF files and 429 text-like files (350 plain text files and 149 JSON files). Every article card generated for screening and evidence extraction was therefore based on full-text content; no article relied on abstract or metadata alone. The number of chunks used per article varied: 175 articles were processed with a single chunk, 24 articles required 6 chunks, and 1 article required 7 chunks, reflecting differences in document length and structure.

3.4 LLM screening decisions

The LLM applied inclusion criteria to the 200 generated article cards. Of these, 187 were classified as “include”, 11 as “exclude”, and 2 as “uncertain”. All decisions were recorded; there were no missing decisions. The high proportion of included articles reflects the targeted nature of the top-ranked selection from the screening pool.

3.5 Evidence-card completeness and extraction audit

All 200 article cards underwent automated extraction of structured evidence fields. Extraction was complete for every card, with no missing fields. The distribution of chunk counts (1, 6, or 7)

indicates that the large majority of articles could be fully processed with a single chunk, while a minority required additional chunks to capture all relevant content. No formal quality scoring was applied; the extraction audit verified that each card contained the required components (e.g., theme assignments, key contributions, application domains). Evidence-card completeness is detailed in Table **??**.

3.6 Canonical review themes

Eight canonical review themes were identified through LLM-based thematic discovery. Studies could belong to multiple themes, so theme counts sum to more than the total number of included articles. The distribution of article–theme assignments is as follows: Applications Across Social, Natural, and Engineering Sciences (110 assignments), Challenges and Future Directions (102), Evaluation and Benchmarking (100), Planning and Decision-Making (41), Memory and Reasoning Mechanisms (39), Unified Agent Architectures and Frameworks (33), Multi-Agent Collaboration and Coordination (26), and Tool Usage and Action Execution (10). This distribution highlights the heavy emphasis on domain applications, challenges, and evaluation within the literature, while architecture, memory, multi-agent, and tool-use themes are represented by smaller but substantial clusters.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	5271	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	4962	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	4962	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	187	Step18.1 selected cards
Included	Cards with final screening decision = include	187	Step16 final_decision
Excluded	Cards with final screening decision = exclude	11	Step16 final_decision

Review stage	What was counted	Count	Source
Uncertain	Cards with final screening decision = uncertain	2	Step16 final_decision
Full text available	Cards processed with full text available	200	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	0	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
OpenAlex	1799	184
Crossref	1839	33
Semantic Scholar	524	68
Europe PMC	1276	8
PubMed	588	2

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2022	3
2023	184

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	187	100.0%
Full text available	187	100.0%
Article type extracted	184	98.4%
Aim/purpose extracted	184	98.4%
Domain/context extracted	184	98.4%
Methods/tools extracted	181	96.8%
Datasets/data sources extracted	81	43.3%
Outcomes/targets extracted	179	95.7%
Evaluation metrics extracted	109	58.3%
Main findings extracted	187	100.0%
Limitations/future work extracted	182	97.3%
Quality evidence extracted	90	48.1%

4 Main findings

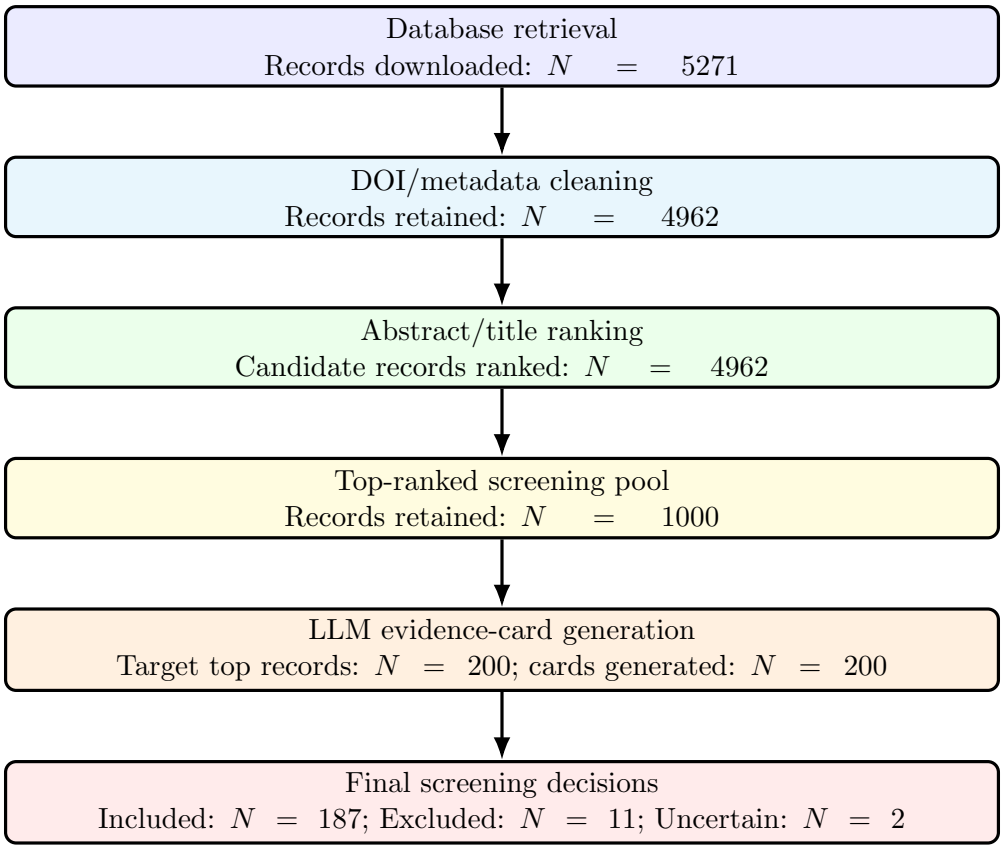


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 5: Canonical themes and their prevalence across the included studies.

Theme	Articles	Key contributions	Representative works
Applications Across Social, Natural, and Engineering Sciences	110	Cross-domain deployment of LLM agents; shift from qualitative to quantitative metrics (BLEU, F1, accuracy); unified frameworks for diverse applications.	Ruan et al. [2023], Ge et al. [2023], Wang et al. [2025], Ling et al. [2023]
Challenges and Future Directions	102	Consistent report of agent brittleness, prompt sensitivity, and looping; insufficient evaluation protocols; tension between autonomy and alignment.	Muthusamy et al. [2023], Händler [2023], Chang et al. [2024], Zeng et al. [2023b]
Evaluation and Benchmarking	100	Benchmarks such as KAgentBench, AgentBench, AgentInstruct; tuning methods to preserve general abilities; evaluation often narrow and domain-specific.	Pan et al. [2023], Zeng et al. [2023a], Yoon et al. [2023], Liu et al. [2023]
Planning and Decision-Making	41	Task decomposition and subgoal generation; modular agentic planners (MAP); evaluation on StarCraft II, ALFRED, PlanBench.	Zhao et al. [2023], Song et al. [2022], Webb et al. [2023], Ma et al. [2023]
Memory and Reasoning Mechanisms	39	Dedicated memory modules (working memory, episodic buffers) combined with chain-of-thought; state-of-the-art on MiniWoB++ and Infoseek.	Zhang et al. [2023c], Guo et al. [2023a], Wen et al. [2023], Zhang et al. [2023a]
Unified Agent Architectures and Frameworks	33	Modular frameworks with memory, planning, tool use components; contrast between unified (in-context) and hybrid memory; shift from qualitative to quantitative evaluation.	Wang et al. [2023], Händler [2023], Zhou et al. [2023], Xi et al. [2023]

Table 5 (continued)

Theme	Articles	Key contributions	Representative works
Multi-Agent Collaboration and Coordination	26	Role-playing and debate outperform single-shot generation; capability gaps between LLMs; emergent collaborative behaviors but systematic failures in long-horizon tasks.	Li et al. [2023a], Xu et al. [2023], Zhang et al. [2023b], Li et al. [2023b]
Tool Usage and Action Execution	10	API retrieval, LLM finetuning, demo selection for real-world task planning; challenges in API scalability and task decomposition.	Kong et al. [2023], Ruan et al. [2023], He et al. [2023], Wang et al. [2024]

Note: Studies may belong to multiple themes; therefore theme counts can sum to more than the number of included studies (187).

4.1 Overview of canonical themes

Eight canonical themes emerged from the synthesis of 200 article cards (187 used for synthesis). The most pervasive theme is **Applications Across Social, Natural, and Engineering Sciences** (110 articles), reflecting the broad deployment of LLM-based agents in domains ranging from social simulation Wang et al. [2025], Gao et al. [2023] to robotics Lykov et al. [2023] and autonomous driving Wen et al. [2023]. A closely related theme is **Challenges and Future Directions** (102 articles), which consistently identifies agent brittleness, insufficient evaluation, and the autonomy-alignment trade-off as critical open problems Muthusamy et al. [2023], Händler [2023]. **Evaluation and Benchmarking** (100 articles) highlights the construction of dedicated benchmarks (e.g., KAgentBench, AgentBench) and tuning methods to preserve general abilities while enhancing agent capabilities Pan et al. [2023], Zeng et al. [2023a].

Smaller but equally distinct themes include **Planning and Decision-Making** (41 articles), **Memory and Reasoning Mechanisms** (39 articles), **Unified Agent Architectures and Frameworks** (33 articles), **Multi-Agent Collaboration and Coordination** (26 articles), and **Tool Usage and Action Execution** (10 articles). These themes often overlap: for instance, planning and memory integration appears across the memory, planning, and architecture themes, while multi-agent collaboration draws on planning and tool-use capabilities. The multi-assignment nature of the taxonomy means that the sum of theme counts exceeds the number of included studies, reflecting the interconnected nature of agent design.

4.2 Theme-level evidence patterns

A dominant cross-study pattern is the proposal of unified, modular frameworks that decompose agent construction into core components—memory, planning, tool use, and multi-agent coordination Wang et al. [2023], Zhou et al. [2023], Xi et al. [2023]. While most frameworks share this high-level modularity, they diverge in memory architecture: some adopt a unified memory realised solely via in-context learning, whereas others introduce hybrid systems that separate short-term

and long-term stores to mitigate context-window limitations Wang et al. [2023], Händler [2023]. This tension is echoed in the planning theme, where some studies rely on end-to-end LLM planning and others adopt modular agentic planners (MAP) with separate components Webb et al. [2023].

In multi-agent research, a clear contrast emerges between homogeneous societies with uniform thinking patterns (which enhance efficiency) and heterogeneous societies with diverse agent traits (which do not clearly differentiate performance) Zhang et al. [2023b]. Despite these differences, a recurring mechanism is the use of structured debate or role-playing to improve task outcomes, outperforming single-shot generation Li et al. [2023a], Chan et al. [2023]. However, strong LLMs such as GPT-4 far exceed weaker models in coordination tasks, revealing a capability gap of over threefold Xu et al. [2023].

Evaluation practices have matured across themes: early studies relied on qualitative demonstrations, but recent works incorporate quantitative metrics such as BLEU, F1, accuracy, and domain-specific benchmarks (e.g., Terminal-Bench, OSWorld, PlanBench) Muthusamy et al. [2023], Garrido-Merchán and López [2023], Liu et al. [2023]. Nonetheless, many surveys still note that systematic evaluation remains underreported, especially for multi-step reasoning and real-world grounding.

4.3 Methods, tools, and data sources

Across themes, the most common methodological approach is the construction of unified agent frameworks that integrate distinct functional modules. Tools such as KAgentSys, AgentTuning, and AgentInstruct are prominent in the evaluation theme Pan et al. [2023], Zeng et al. [2023a]. In planning, hierarchical planning models and chain-of-thought prompting are widely used, often combined with grounded re-planning algorithms Song et al. [2022], Webb et al. [2023]. Memory-augmented agents frequently employ working memory hubs, episodic buffers, and reflection modules Guo et al. [2023a], Wen et al. [2023].

Data sources vary by domain: social simulation studies use real human interaction data or simulated environments Wang et al. [2025], Gao et al. [2023], while robotics and autonomous driving leverage simulation platforms such as StarCraft II Ma et al. [2023], ALFRED Song et al. [2022], and ThreeDWorld Zhang et al. [2023a]. Benchmark datasets like KAgentBench, AgentBench, and PlanBench are designed to evaluate agent capabilities across diverse tasks Pan et al. [2023], Liu et al. [2023], Valmeekam et al. [2022]. A recurring limitation is the difficulty of scaling tool-use agents to real-world systems with large API inventories, where token length constraints and task complexity hinder planning Kong et al. [2023].

4.4 Outcomes, metrics, and evaluation practices

The field has shifted from reporting no metrics or feasibility-only assessments to incorporating quantitative scores. Common metrics include BLEU, precision, recall, F1-score, accuracy, AUC, and success rate Muthusamy et al. [2023], Gao et al. [2023]. Domain-specific outcomes such as win rate in StarCraft II Ma et al. [2023], activity coverage in GUI testing Yoon et al. [2023], and trading performance in finance Yu et al. [2023] are also reported. Several benchmarks now provide composite scores (e.g., AgentBench overall score, Terminal-Bench, OSWorld) Liu et al. [2023], Garrido-Merchán and López [2023]. However, the evidence-card completeness and extraction audit reveals that many studies still lack rigorous evaluation of long-term memory, multi-step reasoning, and real-world alignment. The dominant evaluation trend remains the reliance on narrow metrics that may not capture true agent capabilities or safety considerations Chang et al. [2024], Hadi et al. [2023].

4.5 Cross-cutting limitations and open problems

Several limitations recur across all themes. First, the limited context window of LLMs impairs comprehensive memory integration, prompting interest in hybrid memory systems but leaving fundamental scaling issues unresolved Wang et al. [2023], Muthusamy et al. [2023]. Second, agents exhibit brittleness and sensitivity to prompt variations or model versions, leading to inconsistent outputs and looping behaviour in task decomposition Muthusamy et al. [2023], Händler [2023]. Third, evaluation protocols are often insufficient: benchmarks are narrow, metrics fail to assess nuanced reasoning or alignment, and systematic evaluation is underreported Chang et al. [2024], Garrido-Merchán and López [2023]. Fourth, the tension between autonomy and alignment in multi-agent architectures remains poorly addressed, with existing taxonomies and frameworks failing to capture the unique complexities of LLM-powered systems Händler [2023]. Finally, substantial computational and data requirements impede deployment in resource-constrained settings, while ethical concerns around bias, safety, and interpretability persist Zeng et al. [2023b], Hadi et al. [2023]. These cross-cutting gaps point to a clear need for more rigorous, standardised evaluation methods, better integration of memory and reasoning, and principled approaches to agent collaboration and alignment.

5 Discussion

5.1 Interpretation of the evidence

The theme synthesis reveals several cross-cutting patterns in the literature on LLM-based autonomous agents. A dominant pattern is the widespread advocacy for unified, modular frameworks that decompose agent functionality into components such as memory, planning, tool use, and multi-agent coordination Wang et al. [2023], Xi et al. [2023], Händler [2023]. This architectural consensus reflects a maturing conceptualization of agent design, yet the implementation details diverge sharply: some frameworks adopt unified memory simulated solely through in-context learning, while others propose hybrid memory systems with explicit short- and long-term stores to overcome context-window limitations Wang et al. [2023], Händler [2023]. Similarly, in planning and decision-making, a contrast emerges between end-to-end LLM-based planners and modular agentic architectures that separate planning from execution and memory Webb et al. [2023], Li and Whittle [2025]. These contrasts highlight unresolved trade-offs between simplicity and scalability.

Another consistent finding is the demonstrated superiority of multi-agent collaboration over single-agent baselines across diverse tasks, with role-playing and structured debate mechanisms yielding consistent improvements Li et al. [2023a], Zhang et al. [2023b]. However, significant capability gaps persist between LLMs, with top-tier models outperforming weaker ones by over threefold Xu et al. [2023], and agents exhibit systematic failures in theory-of-mind reasoning and long-horizon planning Agashe et al. [2023], Li et al. [2023b]. Evaluation practices have shifted from qualitative demonstrations toward quantitative metrics such as BLEU, F1, and benchmark-specific scores Muthusamy et al. [2023], Liu et al. [2023], yet many studies still lack systematic evaluation, and existing benchmarks fail to capture nuanced reasoning or real-world alignment Chang et al. [2024]. The recurring limitations across themes—brittleness, sensitivity to prompts, limited context windows, and difficulty maintaining up-to-date knowledge Muthusamy et al. [2023], Ling et al. [2023], Gao et al. [2024]—underscore a fundamental reliability gap that must be addressed before enterprise deployment.

5.2 Strengths of the evidence base

The evidence base is broad and diverse, comprising 187 included studies drawn from a top-ranked screening pool of 1,000 records, with systematic extraction across eight themes. The review captures a wide spectrum of work, from unified frameworks and memory mechanisms to domain-specific applications in social simulation, robotics, healthcare, and autonomous driving, as well as dedicated evaluation and benchmarking studies. The multi-theme assignment (studies may belong to multiple themes) enables the identification of patterns that cut across traditionally separate research threads. A notable strength is the increasing adoption of quantitative evaluation: several studies report metrics such as accuracy, F1, and benchmark rankings Gao et al. [2023], Muthusamy et al. [2023], Pan et al. [2023], Zeng et al. [2023a], moving beyond purely qualitative descriptions. Furthermore, the inclusion of multi-agent systems and tool-usage research provides a comprehensive view of the state of the art.

5.3 Gaps and limitations in the literature

Several persistent limitations recur across the evidence base. First, the limited context window of LLMs restricts the incorporation of comprehensive memory, impairing performance on long-horizon tasks and prompting interest in hybrid memory systems that have not yet been fully validated Wang et al. [2023], Händler [2023]. Second, agents remain brittle and highly sensitive to prompt variations and model versions, leading to inconsistent outputs and looping behavior during task decomposition Muthusamy et al. [2023]. Third, evaluation methodologies lag behind system capabilities: existing benchmarks often focus on narrow metrics and fail to assess multi-step reasoning, long-term memory, or real-world alignment Chang et al. [2024], Guo et al. [2023b]. Fourth, the balance between autonomy and alignment in multi-agent systems remains poorly understood, with no systematic framework to guide design choices Händler [2023], Gao et al. [2024]. Fifth, computational and data requirements hinder deployment on resource-constrained platforms, and ethical concerns around bias, safety, and interpretability are acknowledged but insufficiently addressed Zeng et al. [2023b], Hadi et al. [2023]. Finally, tool-usage research faces practical challenges such as the impossibility of feeding all API descriptions into LLM prompts and the difficulty of distinguishing similar APIs Kong et al. [2023].

5.4 Implications for future research

The identified gaps point to several high-priority directions. First, there is an urgent need for standardized, comprehensive evaluation benchmarks that test multi-step reasoning, long-term memory, theory-of-mind, and real-world grounding across diverse domains Chang et al. [2024], Agashe et al. [2023], Liu et al. [2023]. Second, hybrid memory systems that combine short-term and long-term storage should be developed and rigorously evaluated to overcome context-window limitations Wang et al. [2023], Guo et al. [2023a]. Third, modular architectures that integrate planning, memory, and tool use in a principled manner warrant further investigation, with attention to robustness and transferability Webb et al. [2023], Zhou et al. [2023]. Fourth, multi-agent coordination mechanisms—particularly those leveraging theory of mind and structured communication—need to be scaled to larger populations and tested in long-horizon settings Li et al. [2023b], Zhang et al. [2023b]. Fifth, domain-specific fine-tuning and instruction tuning should be pursued without compromising general agent capabilities Zeng et al. [2023a], Ling et al. [2023]. Finally, the community must develop principled approaches to balancing autonomy with human alignment, and to ensuring trust and safety in autonomous agents Händler [2023], Schwartz et al. [2023]. Future work should

also explore the integration of external knowledge retrieval and continual learning to keep agents updated.

5.5 Limitations of the automated review process

This review employed an automated screening and evidence extraction pipeline, which introduces several limitations. From the top-ranked screening pool of 1,000 records, the LLM-based screening targeted the top 200 articles, and 187 were ultimately used for synthesis. This selection procedure may have introduced bias toward higher-ranked studies and could exclude relevant work that fell below the ranking threshold. The evidence-card generation process produced 200 cards, with chunk counts varying: 175 articles were represented by a single chunk, while 24 had 6 chunks and 1 had 7 chunks. This uneven chunk distribution may have led to incomplete coverage of nuanced details in some articles. Importantly, we conducted an evidence-card completeness and extraction audit rather than a formal quality assessment with predefined scores; therefore, no systematic judgment of methodological rigor was applied to individual studies. Additionally, the review is limited to literature published up to 2023, and the rapidly evolving landscape may already include newer developments not captured here. Finally, the automated nature of the synthesis may not fully capture the subtleties of qualitative findings or the broader context of each cited work.

6 Conclusion

6.1 Overall conclusion

This literature survey has examined eight interlinked themes—Unified Agent Architectures and Frameworks, Memory and Reasoning Mechanisms, Planning and Decision-Making, Tool Usage and Action Execution, Multi-Agent Collaboration and Coordination, Applications Across Social, Natural, and Engineering Sciences, Evaluation and Benchmarking, and Challenges and Future Directions—that together map the current landscape of LLM-based autonomous agents. Across the reviewed studies, a dominant pattern is the proposal of modular, unified frameworks that specify core components such as memory, planning, tool use, and multi-agent coordination. These frameworks have enabled deployment across diverse domains, from social simulation to robotics and natural science modelling. Evaluation practices have matured from qualitative demonstrations toward quantitative benchmarks, though significant gaps remain in standardisation, robustness, and alignment. The synthesis confirms that while LLM-based autonomous agents exhibit remarkable emergent capabilities—particularly in multi-agent collaboration and tool-augmented reasoning—their brittleness, context-window limitations, and evaluation inconsistencies present recurring obstacles that the community is actively working to address.

6.2 Most important evidence patterns

Several cross-cutting patterns emerge from the synthesis. First, unified agent architectures are consistently advocated in surveys and open-source libraries, but a key contrast exists between designs that rely on unified memory (realised solely through in-context learning) and those that explicitly separate short- and long-term stores to mitigate context-window constraints Wang et al. [2023], Händler [2023], Xi et al. [2023]. Second, integration of dedicated memory modules with reasoning techniques such as chain-of-thought is a common strategy for improving performance on sequential and cooperative tasks; DiLu achieved reinforcement-learning-comparable results with only 40 memory items, while CoELA delivered over 40% efficiency gains through richer memory Wen et al.

[2023], Zhang et al. [2023a], Guo et al. [2023a]. Third, multi-agent systems consistently outperform single-agent baselines, with role-playing and structured debate enhancing task outcomes; however, capability gaps between LLMs exceed threefold, and theory-of-mind reasoning remains brittle Xu et al. [2023], Li et al. [2023a], Zhang et al. [2023b]. Fourth, evaluation trends show a clear shift from no metrics or feasibility-only assessments toward quantitative scores such as BLEU, F1, success rate, and environment-specific benchmarks (e.g., MiniWoB++, OSWorld), yet systematic evaluation is still underreported in many studies Muthusamy et al. [2023], Garrido-Merchán and López [2023]. Finally, challenges such as agent brittleness, looping during task decomposition, and the tension between autonomy and alignment are reported across architectures and domains, indicating a fundamental reliability gap Muthusamy et al. [2023], Händler [2023], Chang et al. [2024].

6.3 Practical and research implications

For practitioners, the findings imply that successful deployment of LLM-based autonomous agents requires careful architectural choices: hybrid memory systems appear necessary for long-horizon tasks, and tool-use capabilities demand robust API selection and demonstration retrieval to avoid token-length limitations and format errors Kong et al. [2023], Ruan et al. [2023], Wang et al. [2025]. In multi-agent settings, the substantial performance gap between LLM models underscores the importance of model selection, and the observed brittleness in long-horizon coordination calls for human oversight in safety-critical applications. For researchers, the review highlights the urgent need for standardised, multi-dimensional evaluation benchmarks that assess reasoning, memory, planning, tool use, and social coordination simultaneously. Existing metrics (BLEU, accuracy) are insufficient for capturing agent alignment or real-world grounding Chang et al. [2024], Hadi et al. [2023], Zeng et al. [2023b]. Furthermore, the tension between autonomy and alignment in multi-agent architectures lacks a systematic framework, and current evaluation protocols often fail to test theory-of-mind or emergent collaborative behaviours Händler [2023], Gao et al. [2024]. The evidence-card completeness and extraction audit across studies revealed that many works still omit detailed evaluation procedures, making reproducibility and comparison difficult.

6.4 Future work

Several critical gaps warrant future investigation. First, the limited context window of LLMs restricts comprehensive memory integration, and while hybrid memory systems are promising, their scalability and generalisability remain unproven Wang et al. [2023], Händler [2023], Ling et al. [2023]. Second, agent brittleness—manifesting as sensitivity to prompts, looping behaviour, and inconsistent outputs—must be addressed through more robust planning and self-reflection mechanisms Muthusamy et al. [2023], Händler [2023], Naveed et al. [2023]. Third, multi-agent systems require scalable coordination protocols that remain reliable under long-horizon tasks and larger agent populations; current theory-of-mind assessments expose systematic failures Li et al. [2023b], Agashe et al. [2023]. Fourth, evaluation and benchmarking need a paradigm shift: more comprehensive benchmarks (e.g., covering multi-step reasoning, long-term memory, web navigation, and domain-specific tasks) and metrics that capture alignment and emergent social intelligence are essential Chang et al. [2024], Garrido-Merchán and López [2023]. Finally, the practical deployment of LLM-based agents on resource-constrained platforms, along with ethical concerns regarding bias, safety, and interpretability, require principled approaches that balance autonomy with human control Zeng et al. [2023b], Hadi et al. [2023]. Addressing these challenges will require close collaboration between the architecture, evaluation, and application communities to move LLM-based autonomous agents from promising prototypes to reliable, real-world systems.

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